

MoSCoW Prioritization

Tier	Label	Features & Rationale
M	Must Have (Max 3, Non-negotiable)	<i>Rule: If missing, Demo Day fails entirely.</i> 1. Core Interactive Expression System Includes: Touch/gesture-driven basic feedback (tap-to-smile, shake/flip dizziness, tilt eye follow) + mini TV screen display. Why: This is the product's core identity as a companion device. Without interactive expressions, it is no longer an "emotional pet," and the Demo cannot demonstrate its purpose. 2. Stable Device & Standby Operation Includes: Reliable power, screen output, and looped idle animations for standby state. Why: The device must appear "lively" and functional during the Demo. Without stable operation or idle animations, it feels "dead" and fails to showcase its pet-like persona. 3. Basic Environmental Sensing (Temperature & Humidity) Includes: Real-time temperature/humidity readout from on-board sensors, displayed on the screen. Why: Proves your sensor integration works and adds tangible utility beyond just expressions. Without this, you cannot demonstrate environmental awareness in the Demo.
S	Should Have (Important, Not Launch-Blocking)	<i>Rule: Users will notice and care if missing, but the core product still works.</i> 1. Loud sound-triggered "scared" expression (microphone decibel sensing) 2. Local time/weather display via the Xinzhi Weather API 3. One-tap voice interaction mode (Key A + ATOM module) Why: These features enhance personality, utility, and emotional connection but do not block the core companion experience if they encounter minor issues (e.g., API connectivity or voice module setup).
C	Could Have (Nice-to-Have, If Time Allows)	<i>Rule: A bonus feature, not required for Demo Day success.</i> 1. Mobile web remote expression control (long-press Key B → QR code → phone webpage)

Tier	Label	Features & Rationale
		2. Advanced expression variations (e.g., sleepy, excited animations) Why: These are "cool add-ons" that go beyond the core in-person interaction experience. They can be cut if you need to focus on stabilizing the Must/Should-have features.
W	Won't Have (Out of Scope This Semester)	<i>Rule: Can be cut entirely without impacting the Demo.</i> 1. Persistent user data storage (e.g., remembering interaction history) 2. Multi-user cloud-based remote interaction 3. Long-term battery optimization for daily use (beyond basic Demo power needs) 4. Advanced natural language AI for voice chat (beyond basic ATOM interaction)